

Analog Electronics Portfolio

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Part 1: Transimpedance Based Precision Ammeter

Part 2: Bridge Rectifier Circuit

Purpose:

The purpose of this portfolio is to demonstrate my practical experience with a variety of electrical and electronics related concepts, skills and knowledge. This document will demonstrate theory, instrumentation, design, troubleshooting and component knowledge and skill.

Both circuits were built physically on breadboards. The circuits were modeled and (during the design process) simulated in both KiCad and LTspice. I utilized a variety of test equipment including benchtop, handheld and USB based types that would typically be found in a professional setting.

Instrumentation:

For the ammeter project I used the Rigol DP832 3-channel benchtop PSU to power the circuit, an Arduino DUE as a microcontroller, a Siglent SDS802x oscilloscope and a Fluke 17b+ for general measurements

For the rectifier project I used the digilent Analog Discovery 2 USB oscilloscope and function generator and again my 17b+ for general measurements.

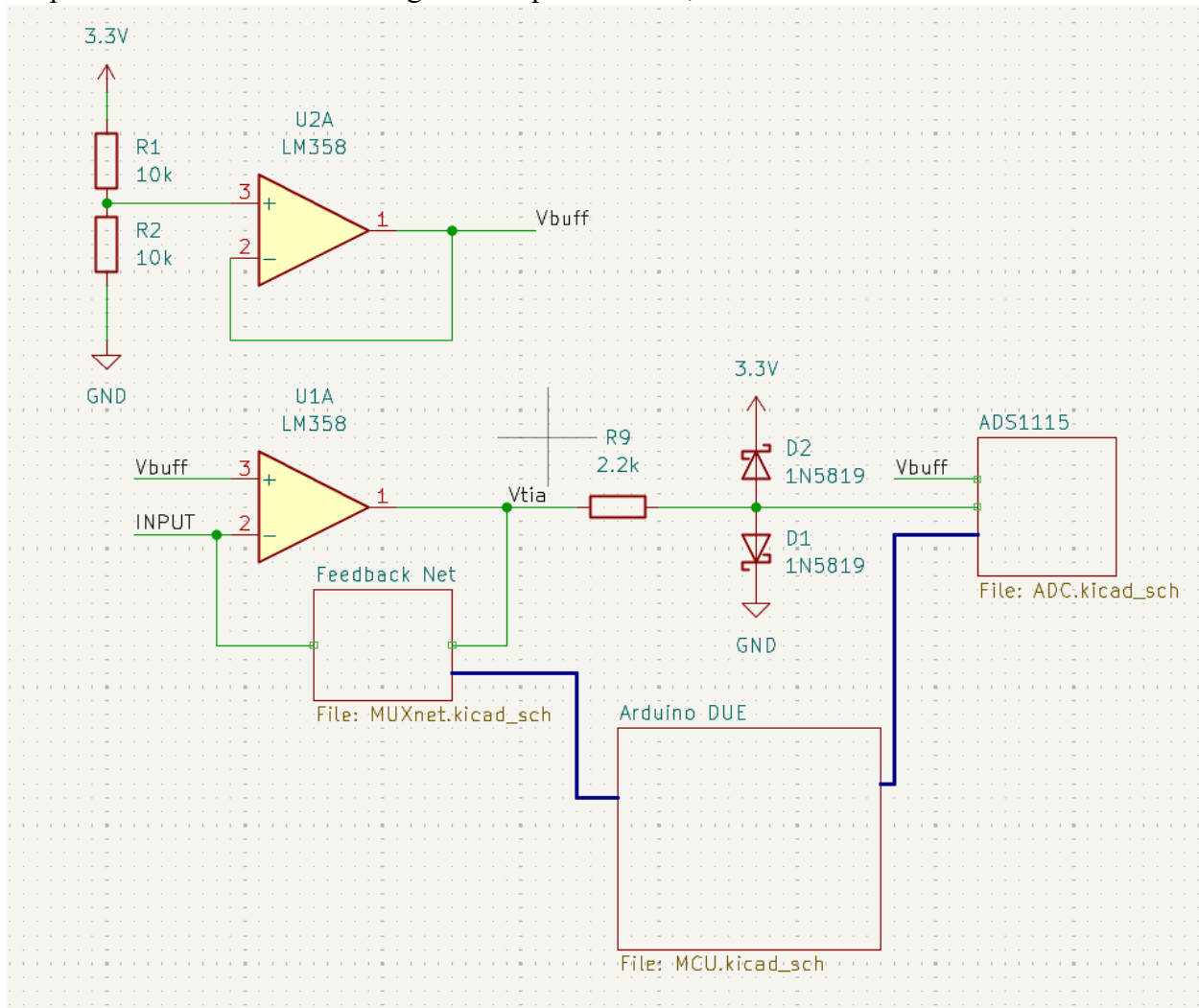
Equipment Used:

- Rigol DP832
- Siglent SDS802x
- Fluke 17b+
- Analog Discovery 2
- Various components and breadboards

Transimpedance Based Ammeter

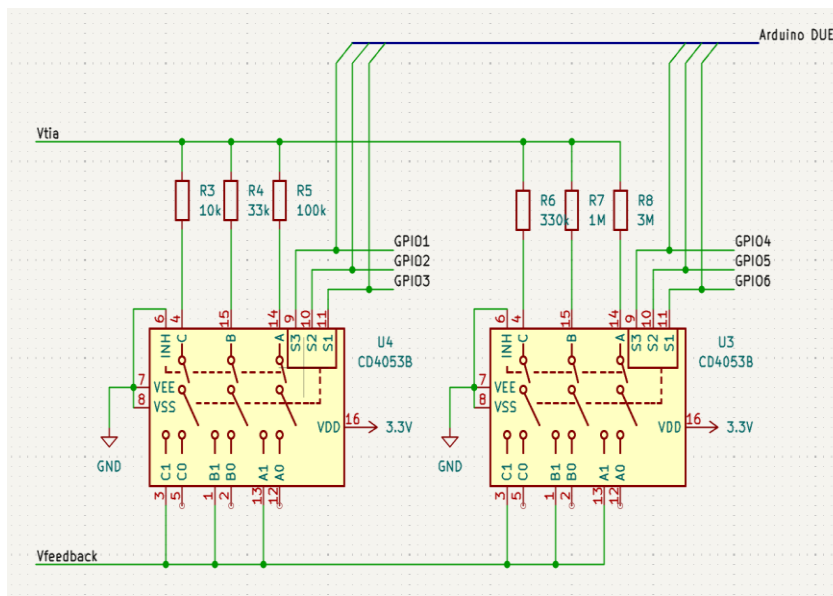
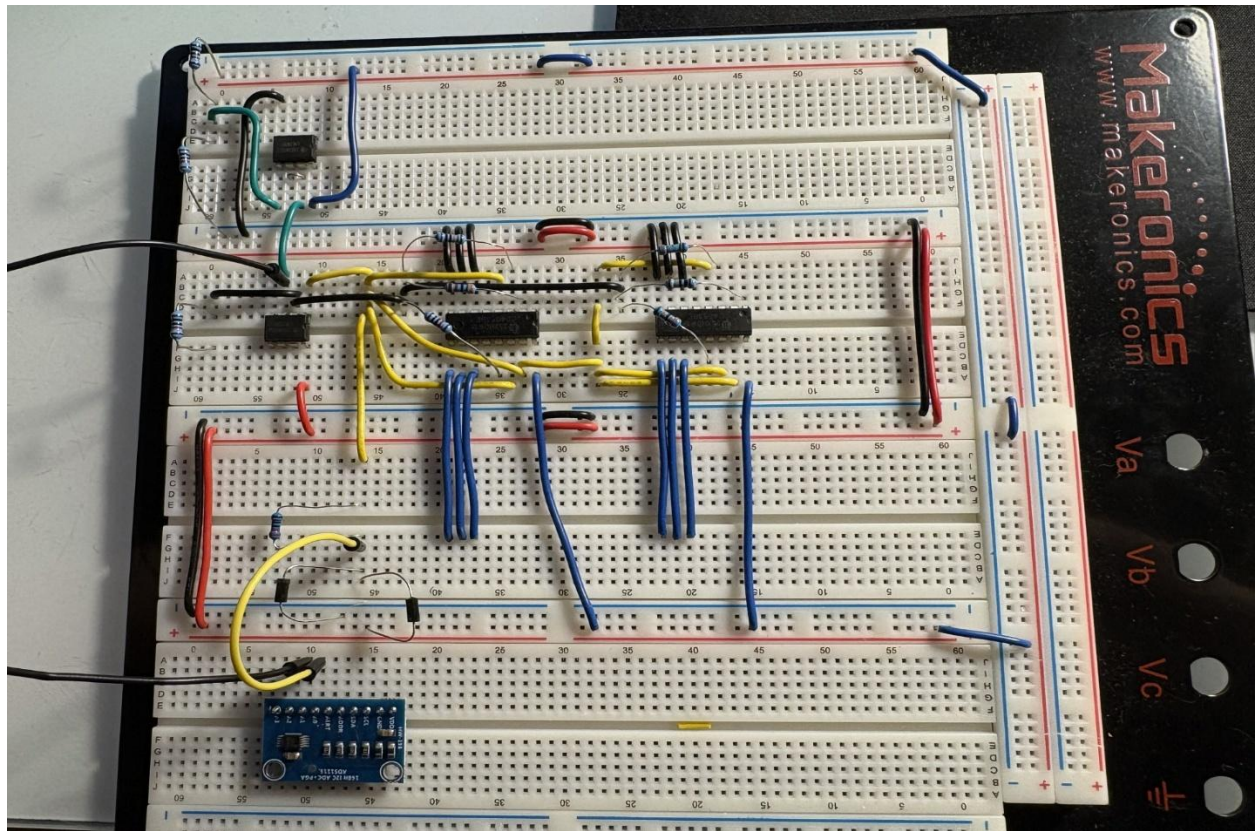
Context:

For very small current measurements, the typical method of using a shunt resistor and measuring voltage is insufficient because of hardware limitations and trade-offs. Instead, a common work-around is a transimpedance based system (transimpedance amplifier or TIA for short) where the measured current is converted into a voltage with an op-amp. It works by outputting a voltage which sources a current through the feedback resistor that cancels out the input current. The circuit I designed is depicted below, modeled in KiCad.



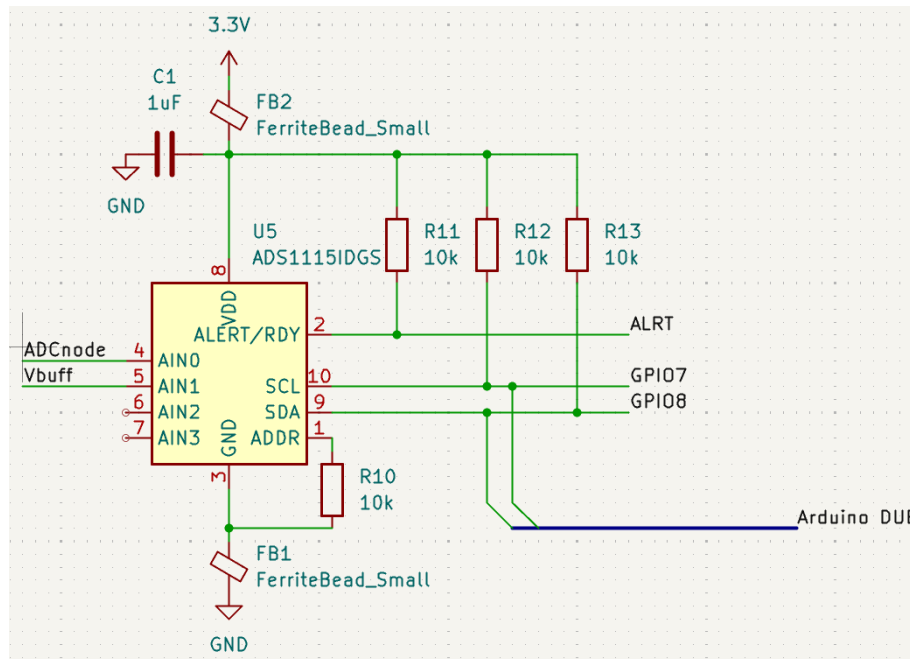
Op-amp U1A is the transimpedance amplifier while U2A is a voltage follower that buffers the 1.65V reference voltage. It's critical to set the reference at 1.65V in this case because neither the Arduino DUE's ADC nor the ADS1115 can tolerate voltage values that are below ground on their pins. The Arduino is programmed such that readings under 1.65V indicate negative current while readings above 1.65V indicate positive currents. The feedback network and ADS1115 sheets will be explored below. A 2.2k Ohm resistor and two Schottky diodes serve as protection for the ADS1115 in case the voltage suddenly jumps too high.

The prototype was built on a triple breadboard with 22 AWG copper hook-up wire as depicted below. The Arduino DUE and the feedback network control bus are not depicted to avoid visual clutter.



One of the more critical sections in this circuit is the feedback resistor network. It uses two CD4053B switch ICs and six resistor values; 10k, 33k, 100k, 330k, 1M and 3M. The Arduino is programmed to automatically increment the current range up or down when the ADC detects that the TIA output crosses a saturation threshold (20% to decrement and 80% to increment.)

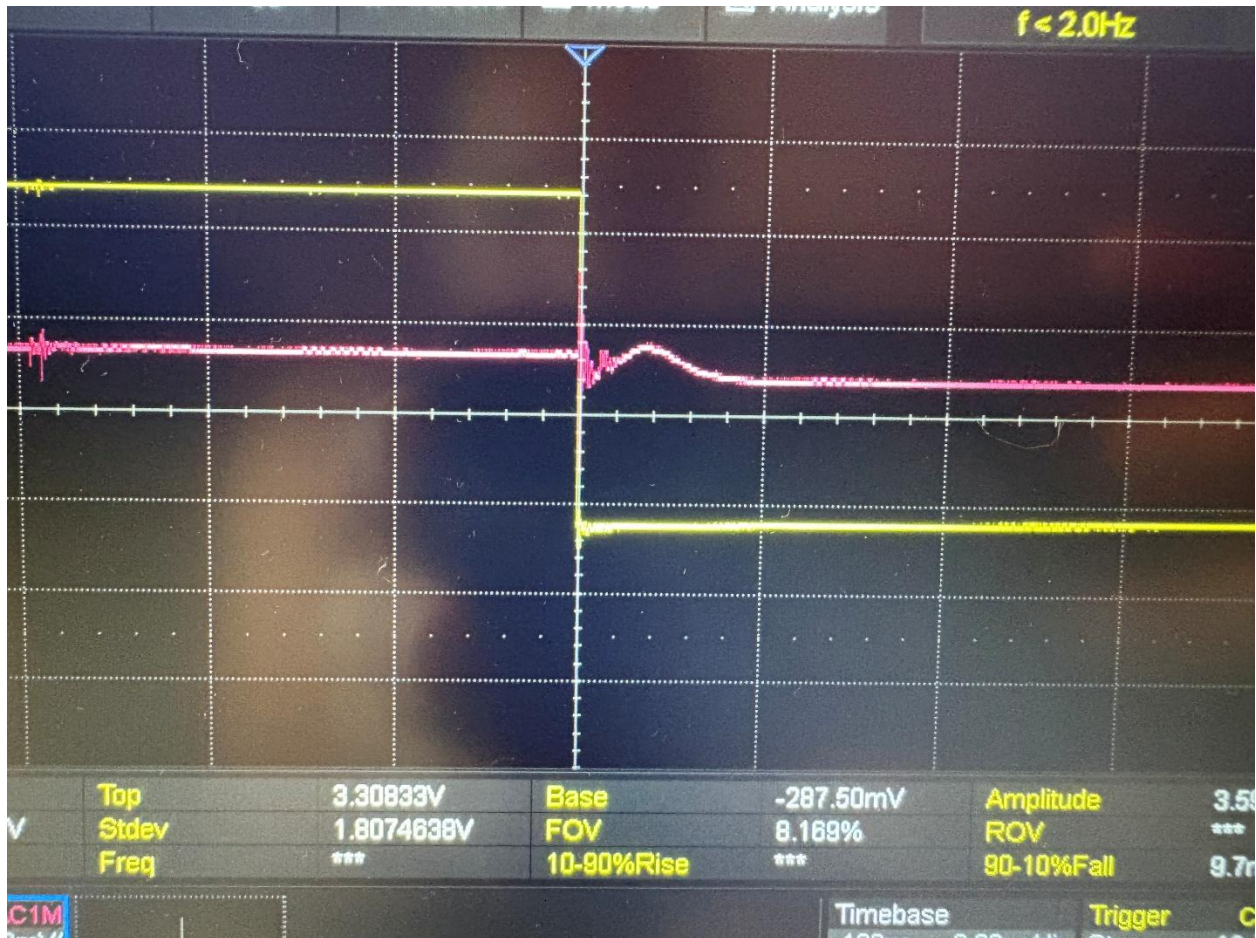
The analog voltage from the TIA is converted to a digital signal with an ADS1115 which, in this case, is soldered to a breakout board. Pull up resistors keep the I2C data and clock pins high which are connected to the corresponding I2C pins on the Arduino.



The Arduino DUE has ADC capabilities but it's limited to 12-bit measurements only. The ADS1115 has a measurement resolution of 16 bits, making it well suited for this application by providing a higher resolution measurement of the signal. To measure negative currents, it requires a differential signal pair. It follows the same convention as the TIA by referencing the measured voltage to the buffered 1.65V.

The feedback network is imperative in this circuit. Being able to switch between feedback resistors gives the TIA six gain ranges which improves the accuracy of the ADC measurements at high and low ranges. Even with a 16-bit ADC if the TIA signal gets too small, it won't map evenly to each resolution step. To combat this, when the current gets smaller, the feedback resistor changes so that the range of the output voltage matches the ADC input voltage. This prevents the ADC resolution from becoming the limiting factor when measuring smaller currents.

The last thing to consider is the settling time of the TIA. Whenever the feedback resistor of the TIA changes, there is a small settling time in which the output swings until it settles on the correct value. The larger the feedback resistor, the longer the settling time. Below is a picture of my Siglent SDS802x. The trigger mode was used to capture the instant the CD4053b switches the feedback resistor.



As mentioned before, the trigger mode was used to capture a snapshot of the exact moment that the logic level changed. Careful inspection shows that the instant that the logic level changes the TIA output exhibits a spike followed by an oscillation until at about a quarter into the time interval division (in this case it's at two microseconds per division.) At around the two microseconds mark it settles to the desired output.

The measure I took to counteract this settling oscillation was to simply measure the settling time for all six resistance values with the oscilloscope and hard-code a delay into the firmware for each current range transition.

Project components:

- Triple breadboard array
- 22 AWG hookup wire
- Arduino DUE microcontroller
- LM358 and CD4053B ICs
- Assorted 1% tolerance resistors
- Schottky diodes

Equipment/software used:

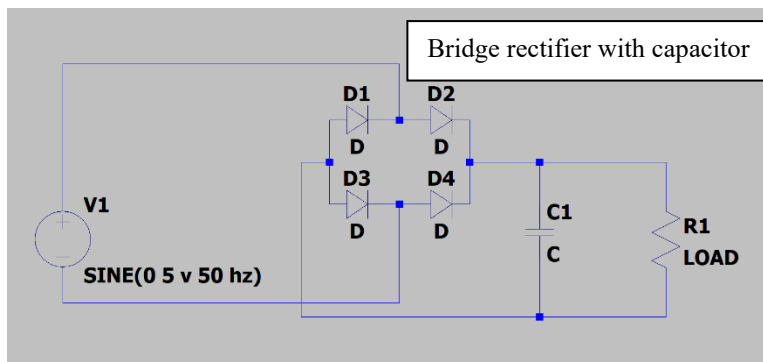
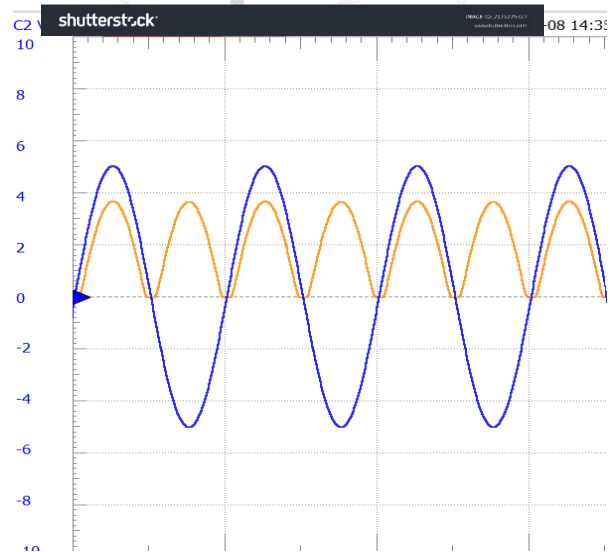
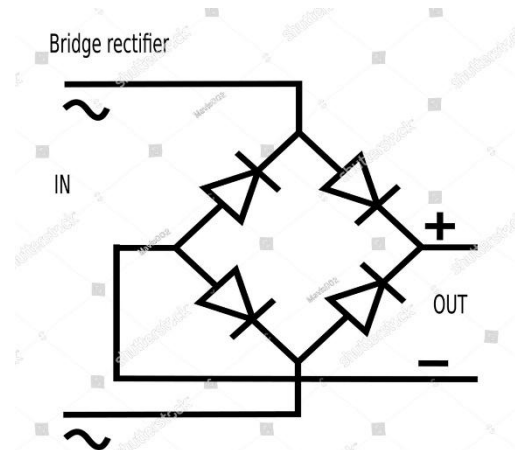
- Rigol DP832 3-channel power supply
- Siglent SDS802x oscilloscope
- Fluke17b+ digital multimeter
- KiCad
- LTspice (I prefer to use it for simulation)
- Arduino IDE

Bridge Rectifier, Filter and Capacitance Multiplier

Context:

Rectifiers are used to convert AC sources and signals into DC. The purpose of this project is to build and test a bridge rectifier with a few different kinds of filters to determine which works best to attenuate ripple and produce a flat DC voltage. The bridge rectifier works by passing an alternating current through a network of diodes. The positive end of the diode network supplies a positive, strobing voltage, eliminating the negative component of the input wave.

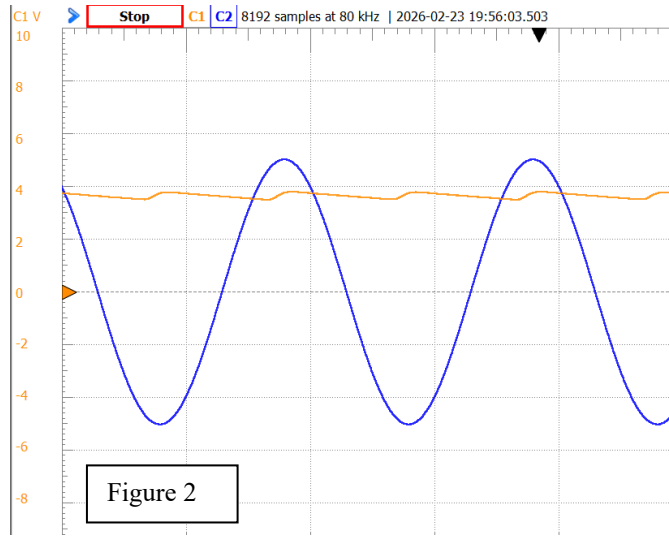
An AC source ($5V_{max}$) enters a network of diodes. A pulsating, positive voltage results, coming from the positive pole of the diode bridge. Figure 1 shows the input voltage (in blue) and the resulting pulsing voltage (in orange).



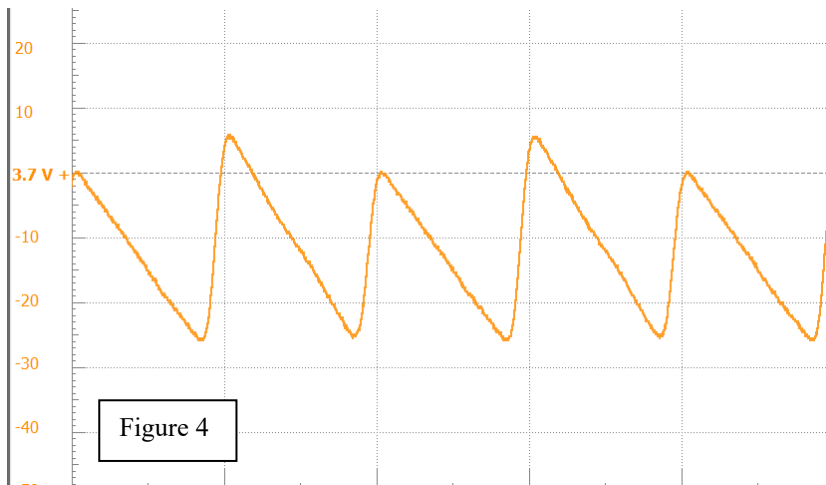
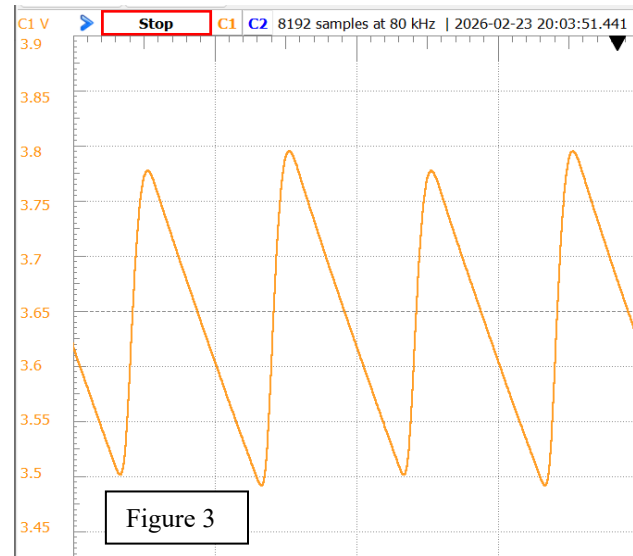
A capacitor added in parallel to the load serves to filter the wave by charging and discharging when V_{in} is negative. The result is a DC voltage with a ripple. The peak to peak range of the ripple can be attenuated by increasing the size of the capacitor.

1. Rectifier with Capacitor Filter:

Figures 2 and 3 depict the ripple when we use a $10\mu\text{F}$ capacitor. The ripple's peak to peak fluctuation is 0.3 volts or 300 millivolts. Figure 3 depicts the ripple when we adjust the range on the oscilloscope. The scale in figure 3 is 50 millivolts per division. It's important to note that figures 2 and 3 depict the waves in both cases at the same frequency. Only the voltage axis is different between the two figures.



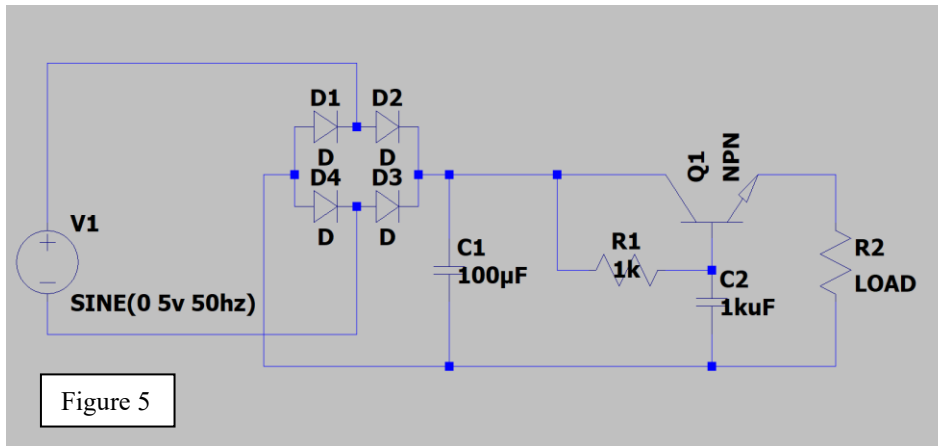
If we replace the $10\mu\text{F}$ capacitor with a $100\mu\text{F}$ one, the filtered signal is attenuated to about 30 millivolts. Below, figure 4 depicts the ripple. In this case, the voltage axis has been scaled to 10 millivolts per division. We can see that the peak-to-peak ripple is ~ 30 millivolts.



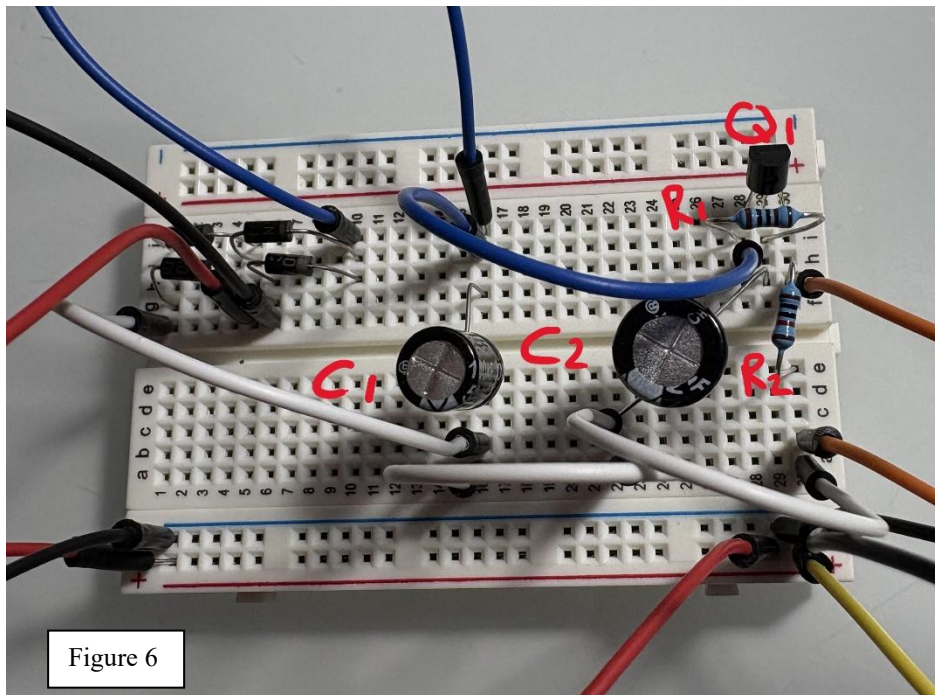
It's also important to note that in both cases, the voltage (after the diode network) has dropped from $5V_{max}$ to $\sim 3.6\text{V}$.

Bridge Rectifier with Capacitor Filter and Capacitance Multiplier:

If we take a NPN transistor and connect an RC network to the base and then connect it in series with our filtered voltage, we can significantly attenuate the ripple. What essentially happens is that the RC network filters the ripple on the base of the transistor which, in turn, attenuates the ripple at the emitter output of the NPN. Figure 5 and 6 depict the circuit and the breadboard prototype respectively.

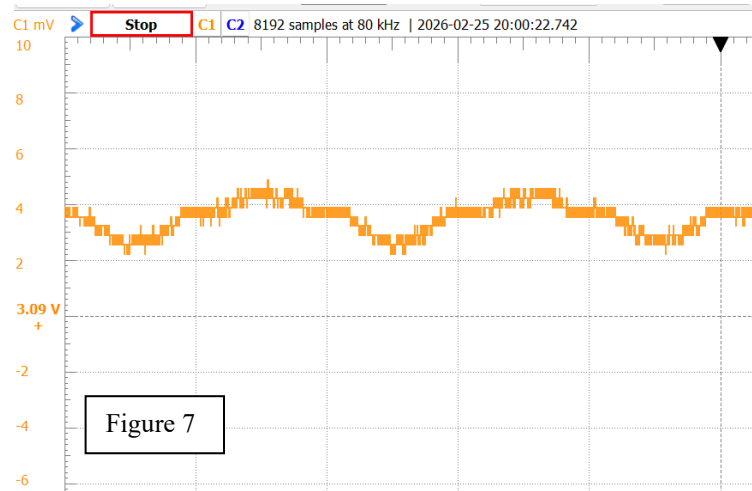


The labeled components in Figure 6 correspond to the labeled circuit elements in Figure 5. On the breadboard, 4 1N4007 diodes were used in the diode network, a 100μF capacitor was used as the first filter, a 1k resistor, a 1kμF capacitor and NPN transistor were used for the capacitance multiplier. The load resistor is arbitrary.



The capacitance multiplier was able to attenuate the ripple to only 2 millivolts peak-to-peak. Figure 7 shows the resulting 'waveform' (at this point it's almost closer to a DC voltage). The voltage axis on the oscilloscope graph is calibrated to 2 millivolts per division.

One important note, however, one drawback of using the capacitance multiplier is that the voltage drops by about $0.6V$ across the transistor. The final voltage across the load is a $\sim 3.1V$ DC signal with only a 2 millivolt ripple.



Parts used:

- Digilent Analog Discovery 2 (For waveform generation and oscilloscope)
- 4 X 1N4007 diodes
- $10\mu F$ electrolytic capacitor
- $100\mu F$ electrolytic capacitor
- $1k\mu F$ electrolytic capacitor
- $1k\Omega$ Resistor